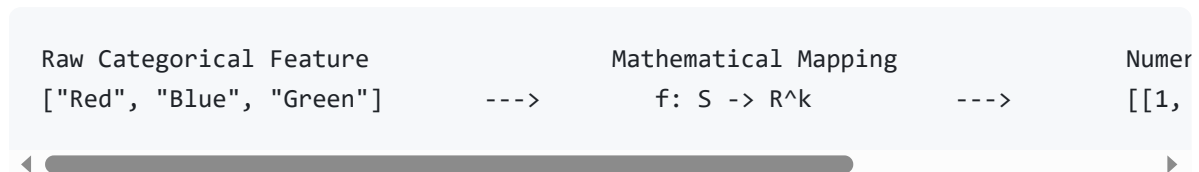


Module 24.01: Feature Engineering — Categorical Encoding

1. The Necessity of Encoding in Mathematical Modeling

Machine Learning algorithms operate on numerical vectors in Euclidean space $\mathbb{R}^d$. Mathematical operations such as matrix multiplications ($\mathbf{X}\mathbf{w}$), Euclidean distance calculations ($|\mathbf{x}_i - \mathbf{x}_j|_2$), and inner products cannot directly ingest non-numeric strings or categorical symbols.



Categorical variables are bifurcated into two foundational mathematical types:

- 1. Nominal Variables:** Categories possess no intrinsic mathematical ordering or hierarchical ranking (e.g., Color: ["Red", "Blue", "Green"], Country: ["India", "USA", "Germany"]).
- 2. Ordinal Variables:** Categories possess a well-defined natural order or magnitude ranking, though intervals between levels may not be uniform (e.g., Education: ["High School", "Bachelors", "Masters", "PhD"], Size: ["S", "M", "L", "XL"]).

2. Encoding Strategies

1. Label / Ordinal Encoding

Ordinal encoding maps each discrete category c_k to an integer scalar $k \in 0, 1, \dots, K - 1$ based on the established rank order:

$$f(c_k) = k \quad \text{where} \quad c_0 < c_1 < \dots < c_{K-1}$$

Critical Warning for Nominal Data: If applied to nominal variables (e.g., Red = 0, Blue = 1, Green = 2), linear and distance-based models mistakenly assume that:

$$\text{Green} > \text{Blue} > \text{Red} \quad \text{and} \quad \text{Green} - \text{Blue} = \text{Blue} - \text{Red}$$

This induces artificial, non-existent geometric constraints into the feature space.

2. One-Hot Encoding (OHE)

One-Hot Encoding converts a categorical feature with K unique levels into K binary indicator columns:

$$\mathbf{x}_{\text{OHE}}^{(i)} = [z_1^{(i)}, z_2^{(i)}, \dots, z_K^{(i)}]^T \in \{0, 1\}^K$$

Where:

$$z_k^{(i)} = \begin{cases} 1 & \text{if sample } i \text{ belongs to category } k \\ 0 & \text{otherwise} \end{cases}$$

Under One-Hot Encoding, every category becomes an orthogonal basis vector in $\mathbb{R}^K$:

- Distance between any two distinct categories $j \neq k$ is identical: $\|\mathbf{e}_j - \mathbf{e}_k\|_2 = \sqrt{1^2 + (-1)^2} = \sqrt{2}$.
- Eliminates any spurious hierarchical ranking.

3. Comparison Matrix of Encoding Techniques

Encoding Method	Target Variable Type	Dimensionality Impact	Risk of Spurious Order	Best Algorithms
Ordinal Encoding	Ordinal only	Preserves d ($1 \rightarrow 1$)	High if applied to nominal	Tree models, Gradient Boosting
One-Hot Encoding	Nominal ($K < 50$)	Expands $d \rightarrow d + K$	None (Orthogonal)	Linear Models, Neural Networks, SVM
Target Encoding	High cardinality nominal	Preserves d ($1 \rightarrow 1$)	Moderate (Target leakage risk)	GBDT, CatBoost, LightGBM
Frequency Encoding	High cardinality nominal	Preserves d ($1 \rightarrow 1$)	Low	Tree-based Ensembles